

Ashes to Analysis – Recreating the VRAF index in Northwest Oregon to assess vegetation recovery after fire

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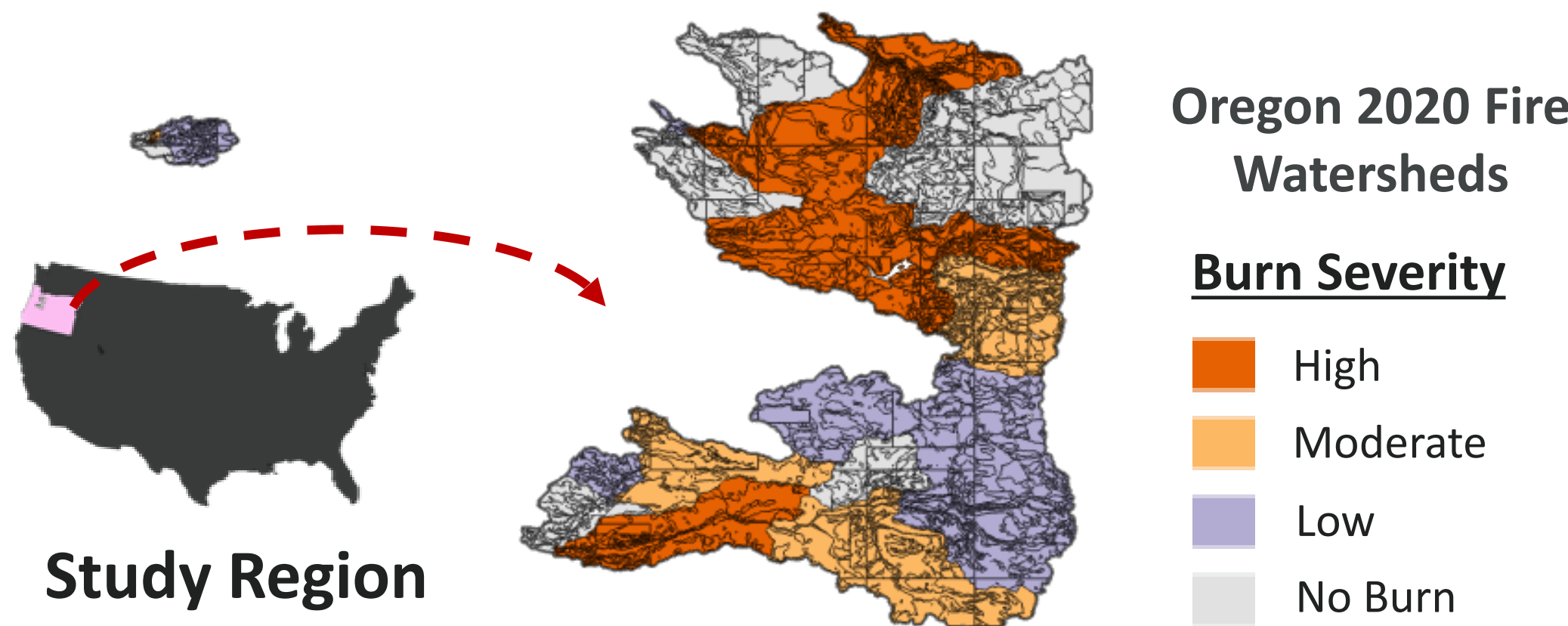
INTRODUCTION

- Large fires like the fire that burned through large portions of Southern California last year and in Oregon in 2020 have a significant effect on the landscapes through which they burn.
- Immediately following a fire event, a landscape is most vulnerable to soil erosion (Bisson et al., 2007), which can have lasting adverse effects for that region.
- The above fact underscores the importance of having geospatial tools to assess vegetation regeneration abilities

OBJECTIVES

- We assessed the recovery capability after fire in the Northwest Oregon region by reconstructing the Vegetation Resilience After Fire (VRAF) index.
- The effectiveness of the index was validated by generating two plots that show the trend of vegetation recovery per year.

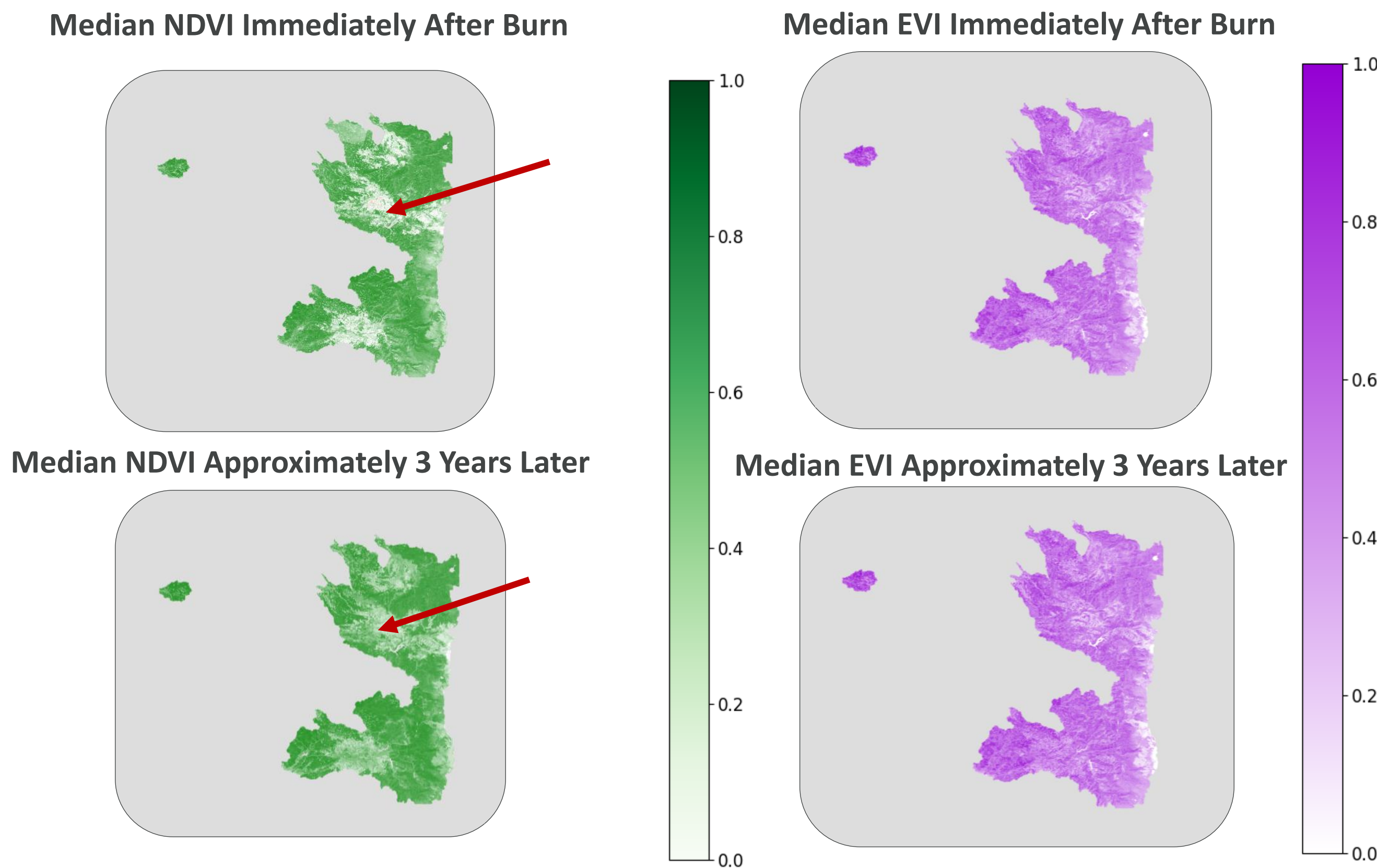
METHODS



Spectral Index Analysis

Formula	Index
Near-Infrared - Red Near-Infrared+Red	NDVI
Near-Infrared - Red Near-Infrared+6 *Red -7.5 *Blue+1	2.5 EVI
Near-Infrared - Mid-Infrared Near-Infrared+Mid-Infrared (Lutz et al., 2011)	NBR
NBRpre-NBRPost (Lutz et al., 2011)	dNBR

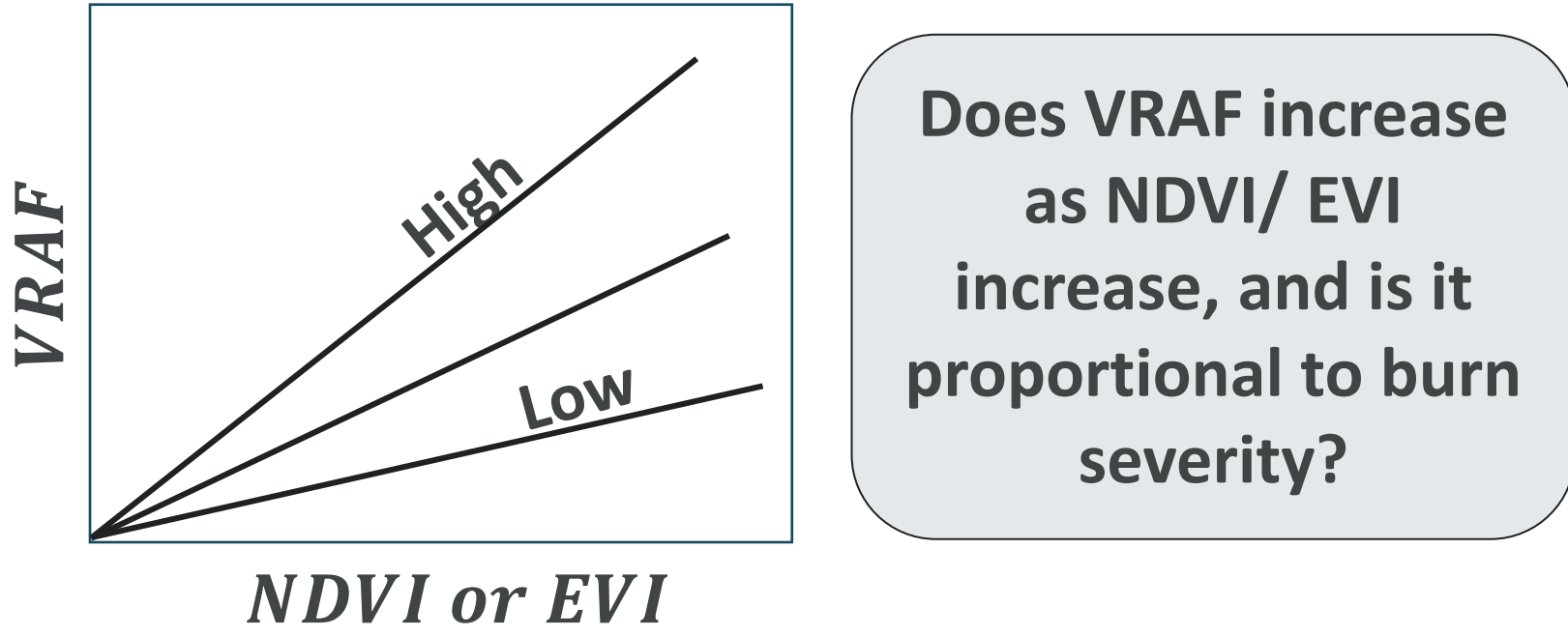
Regional Variability in NDVI and EVI



VRAF

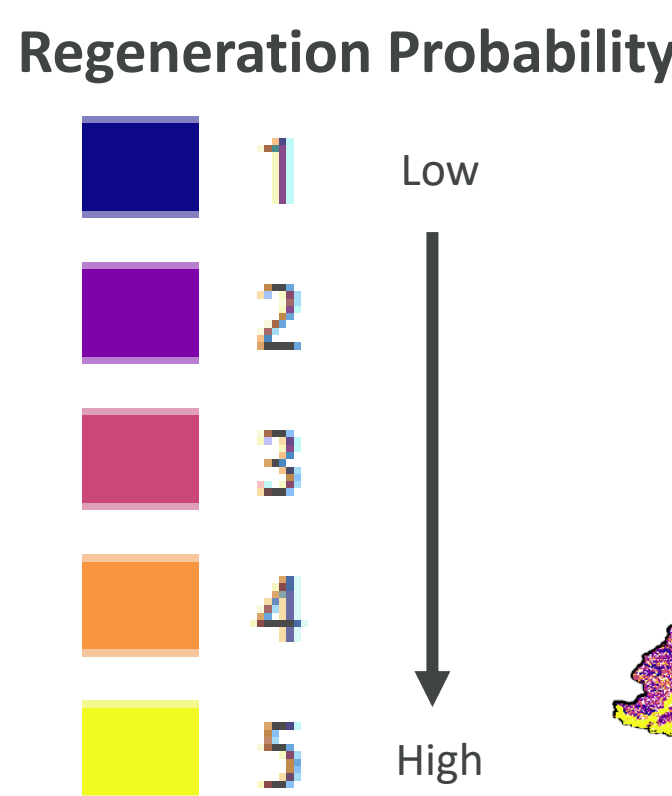
Vegetation Resilience After Fire
 $f(\text{Slope, Aspect, Lithology, Soil, Vegetation and Land Cover})$
$$\text{Resilience Level} = \sum (\text{Pixel VRAF Score}) \times (\text{Weight})$$

VRAF vs Index



RESULTS & DISCUSSION

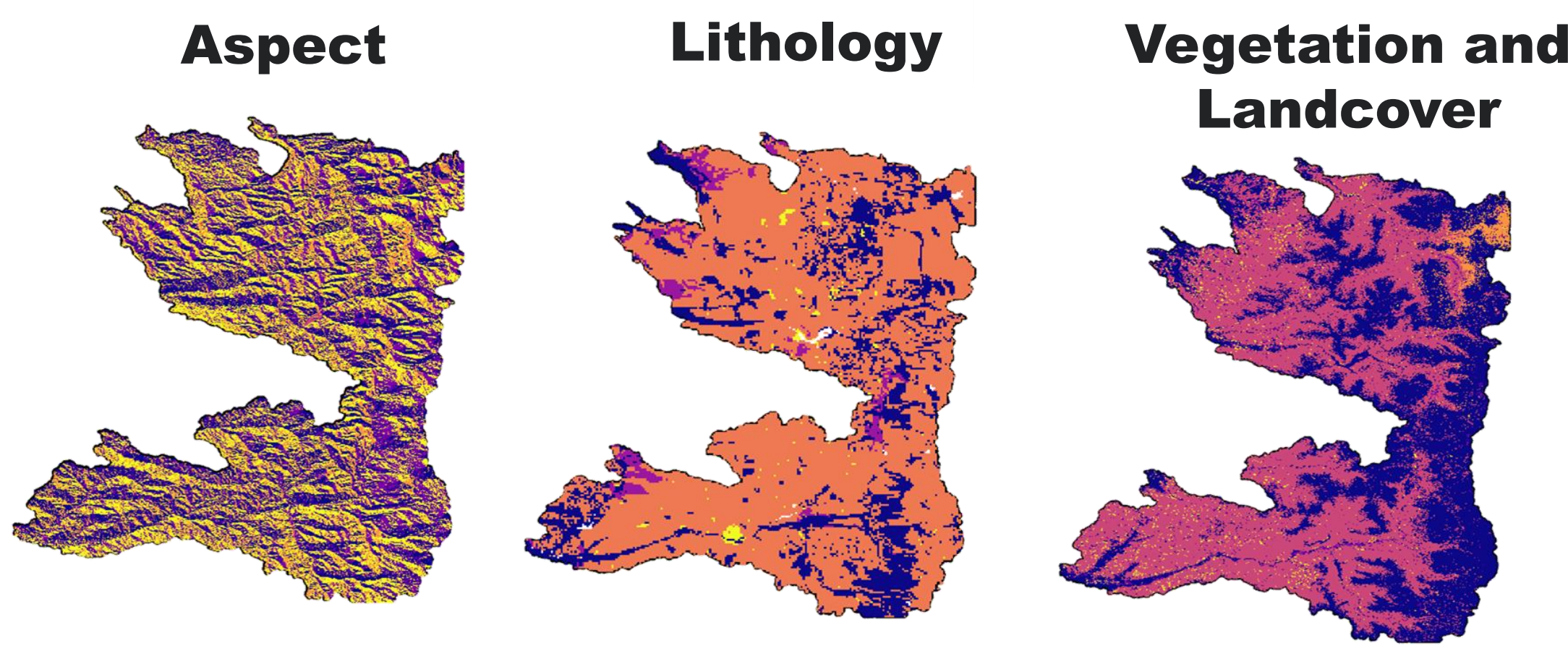
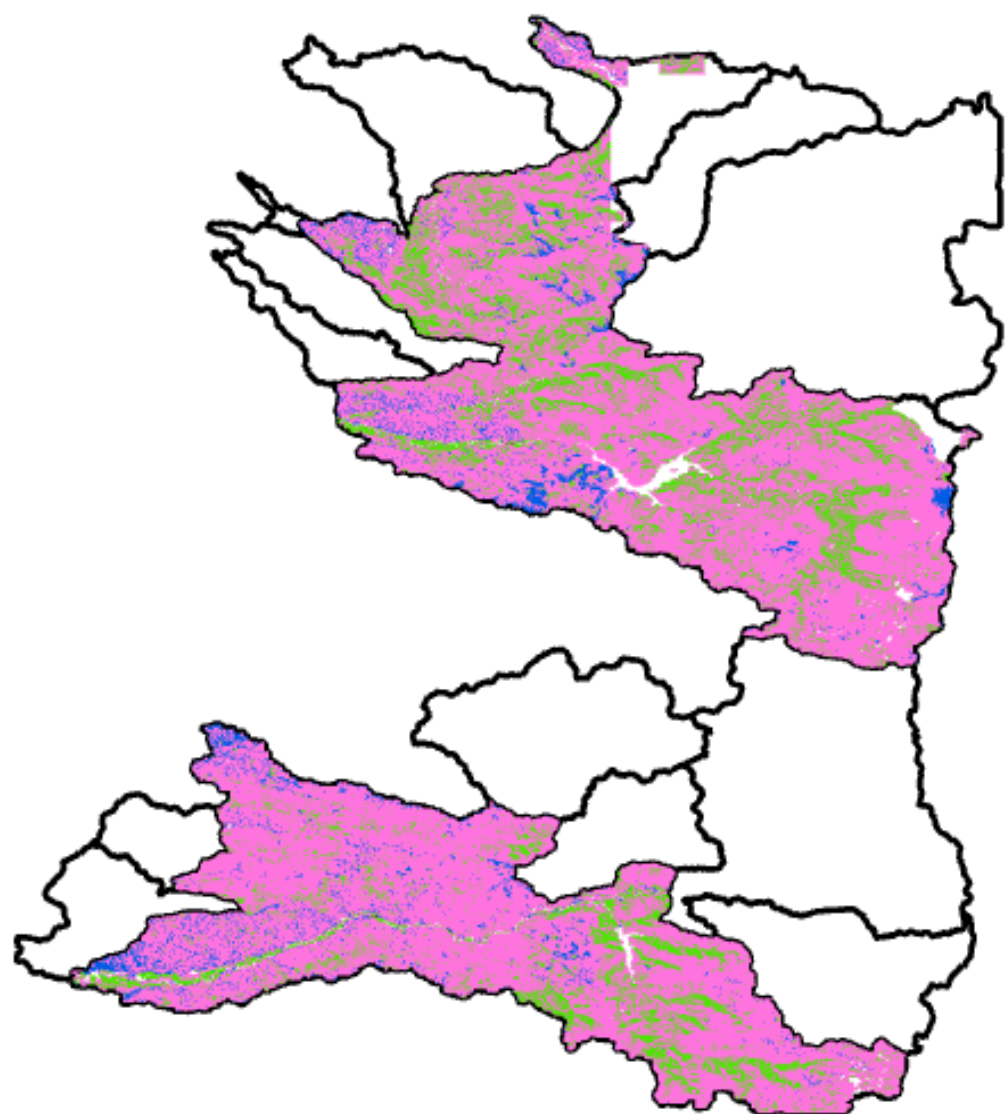
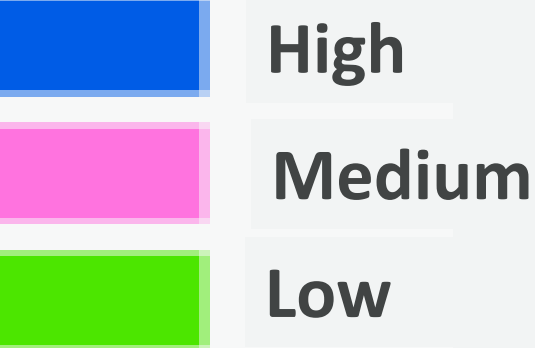
VRAF Score Legend



Slope

Soil

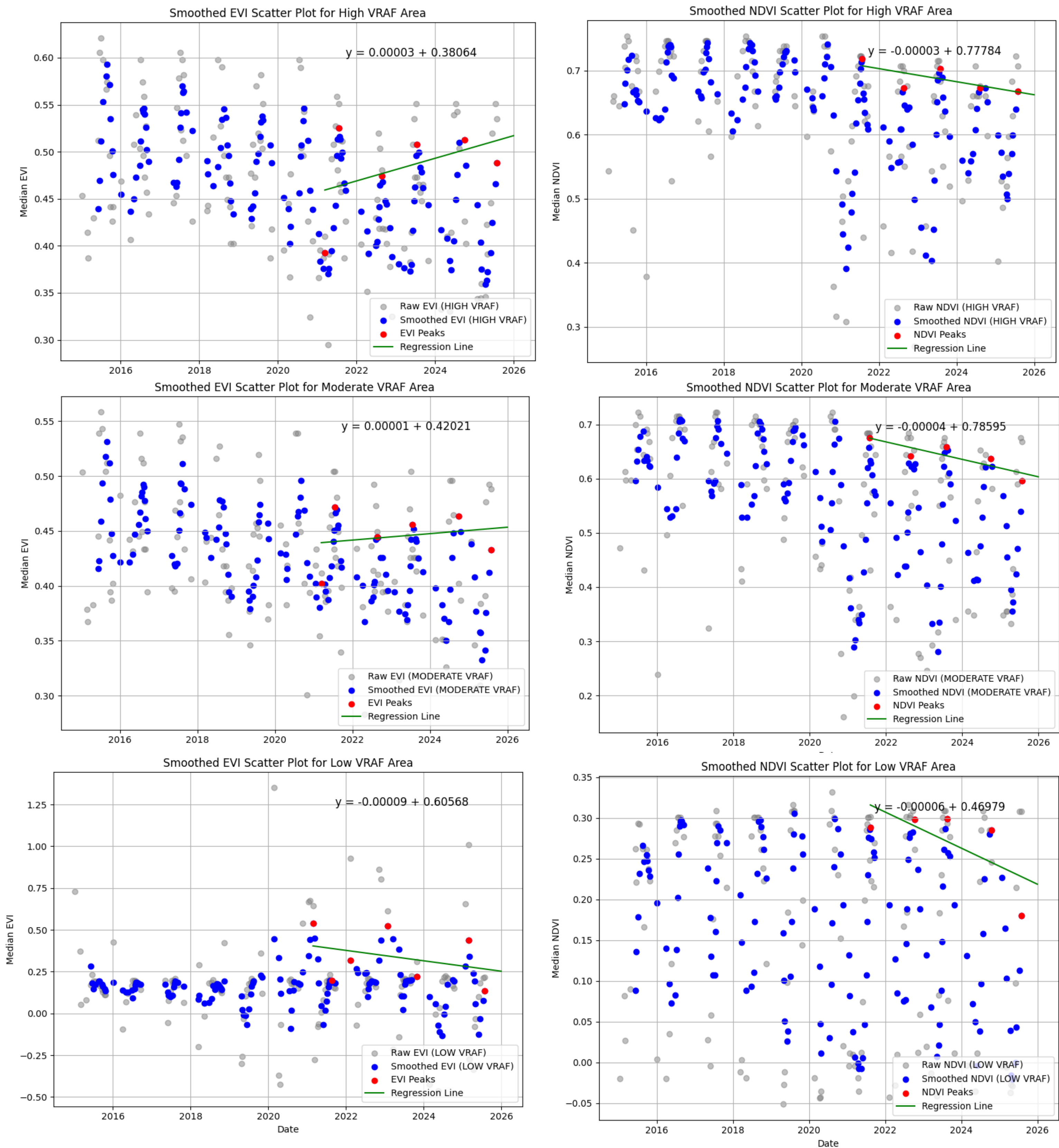
Resilience Level



- The EVI graphs show that higher VRAF levels correspond to greater EVI regeneration rates.
- The NDVI graphs are more difficult to interpret. From an ecological standpoint, the trend may correspond to a shift in forest composition.
- The Low VRAF graph shows a steep decline due to the way that the code detects peaks, but in actuality, we see that the NDVI value is approximately constant across peaks. This is likely because the areas analyzed in this graph are barren regions.

Regeneration Probability	Slope	Soil	Aspect	Lithology	Vegetation and Landcover
1	>40°	D	SW-SE	Surficial Sediments	No Regeneration Possible
2	21-40°	C, C/D	SW-NW	Sedimentary	Regeneration is not Likely
3	11-20°	B/D, A/D	None		Some Regeneration Likely
4	6-10°	B	NE-SE	Volcanic	Moderate Regeneration Likely
5	<5°	A	NE-NW	Plutonic	High Regeneration Likely

We found that a high VRAF value corresponds to a quick recovery NDVI and EVI recovery rate, while a low VRAF value corresponds to low recovery rates.



FUTURE WORK

- Future work includes analyzing the trends of NDVI and EVI Recovery in the entire study region
- Analyzing NDVI and EVI recovery by season
- Working to create a more generalizable data tool that can apply these analyses to any region

WORK CITED

Bisson, M., Fornaciai, A., Coli, A., Mazzarini, F., & Pareschi, M. T. (2008). **The Vegetation Resilience After Fire (VRAF) index: Development, implementation and an illustration from central Italy.** *International Journal of Applied Earth Observation and Geoinformation*, 10, 312–329. <https://doi.org/10.1016/j.jag.2007.12.003>

Lutz, J. A., Key, C. H., Kolden, C. A., Kane, J. T., & van Wagtenonk, J. W.. (2011). **Fire frequency, area burned, and severity: A quantitative approach to defining a normal fire year.** *Fire Ecology*, 7(2), 51–65. <https://doi.org/10.4996/fireecology.0702051>

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